

CPSC-354 Programming Languages (Fall 2026)
 Quiz 1 — Natural Number Game (NNG) (*Mock solutions*)

Time: 30 minutes

Total points: 10

Name	Definition
<code>one_eq_succ_zero</code>	$1 = S(0)$
<code>two_eq_succ_one</code>	$2 = S(1)$
<code>three_eq_succ_two</code>	$3 = S(2)$
<code>four_eq_succ_three</code>	$4 = S(3)$
<code>add_zero</code>	$a + 0 = a$
<code>add_succ</code>	$a + S(b) = S(a + b)$
<code>succ_add</code>	$S(a) + b = S(a + b)$

Figure 1: Names and Definitions

1. Lean to mathematics (5 points).

$$\begin{aligned}
 2 + 2 &= SS0 + SS0 && (\text{two_eq_succ_one}, \text{one_eq_succ_zero}) \\
 &= S(SS0 + S0) && (\text{add_succ}) \\
 &= SS(SS0 + 0) && (\text{add_succ}) \\
 &= SSSS0 && (\text{add_zero}) \\
 &= 4 && (\text{four_eq_succ_three}, \text{three_eq_succ_two}, \\
 &&& \text{two_eq_succ_one}, \text{one_eq_succ_zero})
 \end{aligned}$$

(The Lean proof expands the numeral 4 on the right-hand side down to $SSSS0$ via the numeral lemmas, expands both occurrences of 2 on the left via `two_eq_succ_one` and `one_eq_succ_zero`, then applies `add_succ` twice and `add_zero` once; `rfl` closes because both sides are then $SSSS0$.)

2. Proof by induction (5 points).

We prove $(a + b) + c = (a + c) + b$ by induction on b .

- **Base case** ($b = 0$):

$$(a + 0) + c = a + c = (a + c) + 0,$$

using `add_zero` ($x + 0 = x$) on both sides.

- **Induction step** ($b = Sk$):

Induction hypothesis: $(a + k) + c = (a + c) + k$.

Show: $(a + Sk) + c = (a + c) + Sk$.

$$\begin{aligned}
 (a + Sk) + c &= S(a + k) + c && (\text{add_succ}) \\
 &= S((a + k) + c) && (\text{succ_add}) \\
 &= S((a + c) + k) && \text{induction hypothesis} \\
 &= (a + c) + Sk && (\text{add_succ})
 \end{aligned}$$